

Charles Brackett

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SUMMARY

Skilled Software Engineer with 17 years of varied development experience focusing on mobile development and high quality, reusable code to reduce redundancy during new development and maintenance.

SKILLS

C, C++, C#, Objective-C, Swift, iOS, Git, SQL Server, Visual Studio, Xamarin, Xcode, Troubleshooting

EXPERIENCE

Meridian Staffing Services, LLC, Charlotte, NC

March 2020 – July 2023

Application Developer III; Duke Energy

- Developed RestoreOps, a Swift iOS app that allows damage assessors to evaluate storm damage in the field and upload the data directly to the outage management backend so restoration crews can be dispatched optimally. This replaces a paper process and significantly reduces the time required to restore power to customers in the aftermath of hurricanes and other storms.
- Developed TeamCard, a Swift iOS app used to track timesheets and exceptions for external crews assisting with power restoration after major outage events.
- Maintained OnSight Framework, a Swift framework containing reusable code used by RestoreOps, TeamCard, and other apps within Duke Energy across multiple teams. OnSight Framework significantly reduces the time required for new apps to achieve a minimum viable product by allowing developers to focus on the app's functionality rather than reproducing common UI elements and other logic such as authentication.
- Reviewed all code contributions to OnSight Framework to ensure code quality, reusability, and backwards compatibility.
- Provided feedback and mentored other iOS developers to help them improve their code.

Velocitor Solutions, Charlotte, NC

July 2015 – February 2020

Software Engineer

- Led development of a new framework using C# and Xamarin to maximize code reuse and minimize maintenance. This framework allows C# developers unfamiliar with iOS and Android to write shared business logic for apps that run on both platforms.
- Mobile lead for a generic route accounting app that can either be used out of the box or customized for clients, requiring code to be designed to have minimal duplication of code to allow overriding at a single location. Krispy Kreme currently uses the app in production with no customization and other clients are expected to go to production with customized apps in early 2020.
- Reviewed all contributions to code shared between multiple apps to prevent code issues and to ensure backwards compatibility and override ability.
- Developed new apps for enterprise clients using our C# framework and maintained legacy apps written in C++ for Windows Mobile and Objective-C for iOS.
- Developed and maintained Web APIs for mobile apps using C# and SQL Server.

Kings Mountain International, Inc., Kings Mountain, NC

August 2011 – July 2015

Programmer

- Developed myKMI, an iPad app used during the development phase of KMI's press plates. myKMI allows the user to visualize surface textures on a variety of décors. myKMI is written in Objective-C and presents a gallery of graphics. After downloading graphics, the user can select a texture and décor to load in the Virtual Press Plate view. This custom view uses OpenGL ES to implement a page curl effect using touches to allow the user to peel the texture away from the décor. The Virtual Press Plate view uses tiles at three different resolutions to allow scrolling and zooming within the memory limitations of the first generation iPad. myKMI was available in the App Store and resulted in increased sales of \$800,000 per year.
- Modified Coppermine Photo Gallery to fill the server role of the myKMI project. Wrote a Cron script in PHP to use ImageMagick to convert uploaded décor and texture graphics into tiles at three different resolutions and package them with XML metadata in a ZIP file. Used Restler to provide photo gallery data to the iPad App via JSON.
- Collaborated with coworkers and outside vendors to integrate large industrial printers. Implemented software in C# and C++ to control motion of the machine using Techno and Galil motion controllers and control UV curing

lamps. Motion control software includes the ability to use proximity sensors to detect and adjust print height in order to follow the contours of steel plates at high speed, allowing improved print quality while still printing 8 times faster than the older printer.

- Developed a dust and scratch filter in C# to remove noise from graphics while preserving edge detail. Optimized multithreaded algorithms for performance and to reduce RAM usage in order to scale efficiently for very large graphics (approximately 6 meters x 2 meters at 600 dpi). This software made it possible for the Graphics department to quickly perform tasks that were impossible with existing filters and infeasible to perform manually.

Axiom InSite Solutions, Rock Hill, SC

June 2006 – July 2011

Software Engineer

- Reverse engineered a proprietary print head control library to establish requirements for a replacement. Used Wireshark to analyze TCP and UDP traffic to successfully derive the complete command language and format of the raw print data.
- Designed and wrote a new implementation of a print head control library in C++ to add custom functionality. New functionality includes interlacing support to reduce the impact of issues with individual jets and support for printing with a subset of available jets. Improvements include using LibTIFF to stream the image to allow scaling to very large production graphics (approximately 6 meters x 2 meters at 600 dpi). Includes a wrapper class written in Visual C++ .Net to allow easy use within managed code.
- Integrated the print head control library, a Techno motion controller and an analog I/O board to build a complete control program for a large industrial printer. This software controls the print head and ink system, provides the print head with data, controls motion of the print head over a large area and controls UV curing lamps. The user interface is written in Visual Basic .NET and reusable libraries for the motion control and analog I/O are written in C#. The new printer cut the print time in half when compared to its predecessor.
- Customized and extended LAMP web applications to build and maintain websites for clients.

EDUCATION

Bachelor of Science, Computer Science, North Carolina State University, Raleigh, NC, May 2006

Senior Design Project: Added Ruby support to SlickEdit. Project involved writing a recursive descent parser and lexer in C++ within SlickEdit's framework.

Networking Project: Remote debugging library for the Underwater Robotics Club at NC State. This library uses C/C++ and provides a JNI interface for use with a debugger written in Java.